

DERBY FIELD, NV, USA

WMO: 725805

Lat: 40.068N

Lon: 118.569W

Elev: 1189

StdP: 87.83

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24172

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-13.8	-11.3	-19.2	0.8	-5.7	-16.9	1.0	-4.1	11.7	8.9	9.6	6.5	1.2	110	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	38.0	16.5	36.7	16.1	35.3	15.7	19.0	32.5	17.6	32.2	16.7	31.8	3.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	11.7	24.5	12.2	10.2	23.6	10.3	9.0	22.0	58.6	31.9	53.6	32.1	50.6	31.7	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.1	7.8	DB	-18.3	40.5	3.6	1.7	-21.0	41.7	-23.1	42.7	-25.1	43.6	-27.8	44.8	(4)
(5)			WB	-18.9	21.0	3.5	3.7	-21.5	23.6	-23.5	25.8	-25.5	27.9	-28.1	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	0.5	3.1	6.8	10.1	15.4	20.6	25.2	23.4	18.2	10.9	4.3	-0.4	(6)	
	DBStd	9.42	4.16	3.80	3.82	4.07	4.31	3.98	2.69	2.68	3.84	4.02	4.23	4.54	(7)	
	HDD10.0	1191	294	194	111	49	7	0	0	0	2	37	176	322	(8)	
	HDD18.3	3004	552	427	357	247	113	23	0	2	48	232	422	580	(9)	
	CDD10.0	1758	1	1	12	52	172	319	471	415	247	64	3	0	(10)	
	CDD18.3	529	0	0	0	1	20	92	213	158	44	1	0	0	(11)	
	CDH23.3	10473	0	0	6	79	548	1804	3763	2951	1195	128	0	0	(12)	
	CDH26.7	5743	0	0	0	16	201	928	2306	1706	558	29	0	0	(13)	
(14) Wind	WSAvg	2.9	2.0	2.7	3.2	3.6	3.6	3.7	3.3	2.9	2.7	2.4	2.2	2.0	(14)	
(15) Precipitation	PrecAvg	134	14	13	11	12	14	13	5	7	9	7	11	13	(15)	
	PrecMax	305	46	61	42	55	74	52	38	46	37	43	42	30	(16)	
	PrecMin	80	2	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	49	12	13	10	11	18	14	7	10	10	9	11	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.3	24.0	28.8	33.9	37.9	40.7	39.0	35.7	29.9	21.2	15.1	(19)	
		MCWB	7.8	8.1	9.7	12.0	15.0	16.0	17.6	16.5	15.3	13.4	9.8	7.8	(20)	
	2%	DB	12.0	15.7	21.5	25.6	31.3	35.7	38.9	37.3	33.8	26.9	18.6	12.3	(21)	
		MCWB	6.0	7.3	8.9	10.8	13.6	15.6	17.0	16.0	14.6	12.1	8.8	6.6	(22)	
	5%	DB	10.0	13.6	19.0	23.1	28.9	33.9	37.4	36.0	32.2	24.2	16.2	9.9	(23)	
		MCWB	5.0	6.4	8.0	9.8	12.8	14.9	16.5	15.6	14.2	10.9	7.6	5.1	(24)	
	10%	DB	7.9	11.3	16.8	20.7	26.6	32.2	36.0	34.5	30.2	21.9	13.4	7.6	(25)	
		MCWB	3.9	5.3	7.0	9.0	11.9	14.3	16.3	15.1	13.7	10.0	6.5	3.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.7	11.1	13.3	16.1	18.7	22.0	21.5	18.1	14.6	10.8	8.9	(27)	
		MCDB	12.9	15.9	20.9	25.7	30.6	33.5	36.7	32.8	28.6	26.1	19.2	14.0	(28)	
	2%	WB	6.9	8.2	9.7	11.5	14.4	16.8	19.6	18.3	16.6	12.9	9.5	7.0	(29)	
		MCDB	10.7	14.3	19.4	23.6	28.4	32.6	34.8	30.5	29.0	24.5	16.8	11.3	(30)	
	5%	WB	5.5	6.9	8.5	10.4	13.4	15.6	18.0	16.9	15.3	11.7	8.2	5.4	(31)	
		MCDB	9.0	12.6	17.8	21.7	26.9	31.2	32.8	31.6	28.7	22.1	15.1	9.2	(32)	
	10%	WB	4.2	5.7	7.5	9.3	12.4	14.8	17.0	16.1	14.2	10.5	6.9	3.8	(33)	
		MCDB	7.7	10.9	15.7	19.5	24.8	30.2	32.6	31.4	28.5	20.3	13.0	7.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	14.4	16.9	16.6	16.9	18.9	20.8	21.4	21.2	19.2	16.3	13.3	(35)	
		MCDBR	16.4	17.9	21.7	22.0	21.3	22.0	22.5	23.4	24.2	23.6	20.5	15.4	(36)	
		MCWBR	11.5	11.0	11.9	10.9	8.9	8.9	8.6	9.1	10.6	12.2	12.3	10.7	(37)	
	5% WB	MCDBR	13.9	16.0	19.5	19.8	19.4	20.5	20.4	20.6	21.2	20.7	18.5	14.0	(38)	
		MCWBR	10.1	10.3	11.1	10.0	8.2	8.9	9.0	8.9	9.7	11.1	11.6	10.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.252	0.259	0.270	0.293	0.317	0.310	0.321	0.312	0.293	0.269	0.254	0.250	(40)	
	taud		2.643	2.614	2.579	2.487	2.439	2.486	2.476	2.488	2.536	2.625	2.633	2.635	(41)	
	Ebn at Noon		947	981	997	983	959	962	948	951	955	955	934	922	(42)	
	Edh at Noon		65	77	89	105	112	108	108	103	92	75	64	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.25	3.28	4.64	6.04	7.10	8.16	7.96	7.11	5.84	4.10	2.66	1.89	(44)	
	RadStd		0.34	0.30	0.32	0.37	0.56	0.39	0.34	0.31	0.18	0.27	0.15	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	+88	+46

Nomenclature: See separate page